

1. K-Nearest Neighbors (KNN)

1. What is the main idea behind KNN classification?
 - A. Learn a sequence of decision rules
 - B. Assign a class based on nearby training examples
 - C. Fit a straight line to the data
 - D. Build multiple decision trees
2. In KNN, what does K represent?
 - A. The number of input features
 - B. The number of classes
 - C. The number of neighbors considered
 - D. The number of training iterations
3. A KNN classifier has `n_neighbors=5`. How is a new sample usually classified?
 - A. By the class of the first training sample
 - B. By a vote among its five nearest training samples
 - C. By averaging all training labels
 - D. By selecting five random classes
4. Which Python class is used for KNN classification in scikit-learn?
 - A. `KMeans`
 - B. `KNeighborsClassifier`
 - C. `NearestCentroidRegressor`
 - D. `NeighborTree`
5. Which import statement is appropriate for KNN classification?
 - A. `from sklearn.neighbors import KNeighborsClassifier`
 - B. `from sklearn.tree import KNeighborsClassifier`
 - C. `from sklearn.linear_model import KNeighborsClassifier`
 - D. `from sklearn.ensemble import KNeighborsClassifier`
6. Which method trains a KNN model in scikit-learn?
 - A. `model.learn(X_train)`
 - B. `model.fit(X_train, y_train)`
 - C. `model.predict(X_train, y_train)`
 - D. `model.train(y_train)`
7. Which method predicts labels for new samples?
 - A. `model.fit(X_test)`
 - B. `model.score(y_test)`
 - C. `model.predict(X_test)`
 - D. `model.transform(y_test)`
8. Why is feature scaling often important for KNN?
 - A. KNN requires features to be integers
 - B. Features with larger numerical ranges may dominate distance calculations
 - C. Scaling increases the number of classes
 - D. Scaling removes the need for training data
9. Consider age ranging from 18 to 80 and annual income ranging from 10,000 to 2,000,000. What may happen if KNN uses these features without scaling?
 - A. Age will automatically receive more weight

- B. Income may dominate the distance calculation
 - C. KNN will ignore both features
 - D. The model will become a decision tree
10. Which tool is commonly used to standardize features before KNN?
- A. StandardScaler
 - B. LabelEncoder on the target only
 - C. DecisionTreeClassifier
 - D. confusion_matrix
11. What is a sensible way to combine scaling and KNN in Python?
- A. Pipeline([("scaler", StandardScaler()), ("knn", KNeighborsClassifier())])
 - B. KNeighborsClassifier(StandardScaler())
 - C. StandardScaler(KNeighborsClassifier())
 - D. Pipeline([("knn", KNeighborsClassifier()), ("scaler", StandardScaler())])
12. Why should a scaler be fitted only on the training data?
- A. To reduce the number of classes
 - B. To prevent information from the test set entering model preparation
 - C. To make KNN use Euclidean distance
 - D. To remove all missing values
13. What generally happens when K is set to 1?
- A. The classifier considers the single closest training example
 - B. The classifier uses all training examples
 - C. The classifier becomes a regression model
 - D. The classifier ignores distance
14. A very small K can make KNN particularly sensitive to what?
- A. Feature names
 - B. Noise and unusual training examples
 - C. The order of Python imports
 - D. The number of CPU cores only
15. A very large K may cause which problem?
- A. Predictions may become overly smooth and miss local patterns
 - B. The model memorizes every training label perfectly
 - C. Distance calculations stop working
 - D. The number of input features becomes zero
16. What is the purpose of trying several K values with cross-validation?
- A. To choose a K that performs well on unseen data
 - B. To change the target labels
 - C. To avoid using a test set entirely
 - D. To make all predictions identical
17. For a binary classification problem, why might someone initially try an odd K?
- A. It guarantees perfect accuracy
 - B. It can reduce voting ties
 - C. It eliminates the need for scaling
 - D. It makes KNN train faster in every case
18. What does weights="distance" do in KNeighborsClassifier?
- A. Gives closer neighbors greater influence

- B. Gives every neighbor zero influence
 - C. Weights features according to their names
 - D. Converts classification into regression
19. How does `weights="uniform"` treat the selected neighbors?
- A. All selected neighbors contribute equally to the vote
 - B. Only the closest neighbor votes
 - C. More distant neighbors always vote twice
 - D. Each feature receives the same coefficient
20. Which distance measure is commonly associated with straight-line distance between points?
- A. Euclidean distance
 - B. Gini impurity
 - C. Log loss
 - D. Information gain
21. Which distance measure sums absolute differences across features?
- A. Manhattan distance
 - B. Euclidean distance
 - C. Entropy
 - D. Precision
22. In scikit-learn KNN, what does `metric="manhattan"` change?
- A. The distance used to find neighbors
 - B. The number of target classes
 - C. The training labels
 - D. The evaluation metric returned by score
23. What happens when `n_neighbors` is larger than the number of available training samples?
- A. Prediction can fail because there are too few samples
 - B. KNN creates new training samples automatically
 - C. The model switches to Naive Bayes
 - D. KNN silently removes all features
24. Which type of task is `KNeighborsRegressor` designed for?
- A. Predicting a continuous value
 - B. Grouping data without labels
 - C. Predicting only text labels
 - D. Selecting important features
25. How does basic KNN regression predict a value?
- A. It combines target values of nearby training samples
 - B. It chooses the most common class label
 - C. It fits one global linear equation
 - D. It counts the number of decision tree leaves
26. Which is an appropriate KNN regression problem?
- A. Predicting house price from features of similar houses
 - B. Assigning an email to spam or not spam
 - C. Identifying whether an image contains a cat
 - D. Classifying a tumor as benign or malignant
27. Which is an appropriate KNN classification problem?
- A. Predicting a patient's exact blood pressure

- B. Predicting whether a patient belongs to a risk category
- C. Predicting a product's selling price
- D. Predicting tomorrow's temperature

28. Why can KNN prediction become slow on a very large training set?

- A. It may need to search many stored training samples for neighbors
- B. It retrains a neural network for each sample
- C. It builds a new random forest for each prediction
- D. It must calculate entropy for every feature

29. What does it mean that KNN is an instance-based method?

- A. It uses stored training examples when making predictions
- B. It creates one rule for every class and discards the data
- C. It requires exactly one feature
- D. It trains only on images

30. How can irrelevant features affect KNN?

- A. They can distort which samples appear close
- B. They are always ignored automatically
- C. They increase the number of labels
- D. They turn continuous targets into categories

31. What is the "curse of dimensionality" in the context of KNN?

- A. Distances may become less informative as the number of features grows
- B. Python cannot store more than three features
- C. KNN requires one neighbor per feature
- D. High-dimensional data always gives perfect predictions

32. Two records have identical feature values but different class labels. What difficulty can this create for KNN?

- A. The nearby evidence conflicts
- B. KNN cannot store either record
- C. The number of features doubles
- D. Distance becomes negative

33. Which scikit-learn call returns class probabilities from a fitted KNN classifier?

- A. `model.predict_proba(X_test)`
- B. `model.predict_probability(y_test)`
- C. `model.fit_proba(X_test)`
- D. `model.class_probabilities()`

34. In a three-neighbor uniform-voting KNN model, two neighbors belong to class A and one to class

B. What is the predicted class?

- A. A
- B. B
- C. Both classes are always returned
- D. The prediction cannot be made

35. What is a reasonable first step if one feature is measured in millimeters and another in kilometers?

- A. Scale the features appropriately
- B. Increase K to the number of training samples
- C. Remove the target column
- D. Use training accuracy as the only evaluation

36. What does `train_test_split(X, y, test_size=0.2)` commonly do?
- A. Reserves approximately 20% of samples for testing
 - B. Uses 20 features for testing
 - C. Removes 20% of the classes
 - D. Sets K to 0.2
37. Why should test samples remain separate while selecting K?
- A. To keep the final test evaluation independent
 - B. To make the training set smaller for no reason
 - C. To force Euclidean distance
 - D. To prevent the use of cross-validation
38. Which metric is useful when classifying a rare disease and missing a positive case is especially costly?
- A. Recall for the positive class
 - B. Training set size alone
 - C. Number of features alone
 - D. Prediction time alone
39. What does `model.score(X_test, y_test)` return for `KNeighborsClassifier` by default?
- A. Classification accuracy
 - B. Mean squared error
 - C. Number of neighbors
 - D. Euclidean distance
40. Why can accuracy be misleading on a highly imbalanced classification dataset?
- A. Predicting the majority class can produce high accuracy while missing the minority class
 - B. Accuracy always becomes negative
 - C. KNN cannot use imbalanced data
 - D. Accuracy is defined only for regression
41. A dataset contains a categorical feature such as blood type. What should you consider before applying standard distance-based KNN?
- A. How to encode the category and whether the resulting distance is meaningful
 - B. Converting the target to a floating-point value
 - C. Setting every blood type to the same number
 - D. Increasing the number of test samples
42. Why is assigning arbitrary numbers such as 1, 2, and 3 to unordered categories potentially problematic for KNN?
- A. The numbers imply distances and ordering that may not exist
 - B. KNN accepts only strings
 - C. Numeric features are ignored
 - D. The target must contain three classes
43. What is the likely effect of adding many unrelated random features to a KNN model?
- A. Neighbor selection may become less reliable
 - B. The model must always improve
 - C. K automatically becomes 1
 - D. Scaling becomes impossible
44. What does `GridSearchCV` help you do with a KNN pipeline?
- A. Evaluate candidate hyperparameters using cross-validation
 - B. Create target labels from scratch

- C. Replace the test set with predictions
- D. Draw a decision tree

45. Which parameter would you vary to study how many neighbors influence predictions?

- A. `n_neighbors`
- B. `random_state`
- C. `max_depth`
- D. C

46. What is a limitation of applying ordinary KNN directly to raw, high-resolution images?

- A. Pixel-space distance may not match meaningful visual similarity
- B. Images cannot be converted into arrays
- C. KNN accepts only two input features
- D. KNN always returns continuous values

47. If closer neighbors should matter more, which setting is most relevant?

- A. `weights="distance"`
- B. `weights="uniform"`
- C. `max_depth=3`
- D. `criterion="entropy"`

48. When is KNN especially intuitive as a baseline?

- A. When similar feature profiles are expected to have similar outcomes
- B. When no labeled training examples are available
- C. When every prediction must be a linear equation
- D. When the target has no relationship with the features

49. In `model.fit(X_train, y_train)`, what does `X_train` contain?

- A. Training input features
- B. Predicted probabilities
- C. Test labels
- D. Evaluation scores

50. In `model.fit(X_train, y_train)`, what does `y_train` contain?

- A. Training target values or class labels
- B. Feature scaling parameters only
- C. Test predictions
- D. Distance metrics

2. Naive Bayes

1. What is Naive Bayes primarily used for?
 - A. Supervised classification
 - B. Drawing decision trees
 - C. Clustering without labels
 - D. Sorting a dataset by date
2. What does the “naive” assumption refer to?
 - A. Features are treated as conditionally independent given the class
 - B. The model assumes every class is equally likely in all datasets
 - C. The model uses no training data
 - D. The model always predicts the first class
3. Which theorem provides the foundation for Naive Bayes?
 - A. Bayes’ theorem
 - B. Pythagoras’ theorem
 - C. The central limit theorem only
 - D. The law of large numbers only
4. In a spam classifier, what does the model estimate?
 - A. The probability of a class given features of an email
 - B. Only the number of characters in an email
 - C. The Euclidean distance between all emails
 - D. The depth of a decision tree
5. Which Python class is commonly used for Naive Bayes with continuous features assumed to follow class-specific Gaussian distributions?
 - A. GaussianNB
 - B. MultinomialNB
 - C. BernoulliNB
 - D. KNeighborsClassifier
6. Which Naive Bayes variant is commonly used with word-count features in text classification?
 - A. MultinomialNB
 - B. GaussianNB
 - C. BernoulliNB only for every text dataset
 - D. DecisionTreeRegressor
7. Which variant naturally models binary features such as whether a word appears?
 - A. BernoulliNB
 - B. GaussianNB
 - C. LinearRegression
 - D. RandomForestRegressor
8. Which import statement is appropriate for Gaussian Naive Bayes?
 - A. `from sklearn.naive_bayes import GaussianNB`
 - B. `from sklearn.neighbors import GaussianNB`
 - C. `from sklearn.tree import GaussianNB`
 - D. `from sklearn.metrics import GaussianNB`
9. Which statement creates a Gaussian Naive Bayes model?
 - A. `model = GaussianNB()`
 - B. `model = GaussianNB.fit()`

C. `model = BayesGaussianClassifier`

D. `model = GaussianNB.predict()`

10. What does `model.fit(X_train, y_train)` do?

A. Learns parameters from labeled training data

B. Predicts labels for the test set

C. Calculates only test accuracy

D. Removes duplicate records automatically

11. What does `model.predict(X_test)` return?

A. Predicted class labels

B. A confusion matrix

C. The training dataset

D. A list of feature names

12. What does `model.predict_proba(X_test)` return?

A. Estimated probabilities for each class

B. Only distances between samples

C. A new training set

D. A decision tree diagram

13. In Bayes' theorem, what is a prior probability?

A. The probability of a class before considering the current sample's features

B. The probability that a feature is missing

C. The final prediction for a sample

D. The test accuracy

14. In a dataset where 90% of emails are legitimate, what can the model learn about class priors?

A. Legitimate email is more common in the training data

B. Both classes must have a prior of 0.5

C. Spam is always more probable

D. Priors are unrelated to class frequency

15. What is a likelihood in Naive Bayes?

A. How probable the observed features are under a particular class

B. The number of model parameters

C. The distance to the nearest sample

D. The final accuracy on a test set

16. Which application is a common use of Naive Bayes?

A. Document topic classification

B. Image resizing

C. Database backup

D. Sorting files alphabetically

17. Why is Naive Bayes often a useful baseline for text classification?

A. It can work efficiently with high-dimensional word features

B. It requires no labels

C. It always understands word meaning and context perfectly

D. It needs every document to have the same number of words

18. What does `CountVectorizer` generally produce from text documents?

A. Numeric features representing token counts

B. A decision tree

- C. A list of class predictions
- D. A set of images

19. Which pipeline is suitable for a basic word-count spam classifier?

- A. CountVectorizer() followed by MultinomialNB()
- B. GaussianNB() followed by CountVectorizer()
- C. KNeighborsClassifier() followed by CountVectorizer()
- D. confusion_matrix() followed by MultinomialNB()

20. Why must text be converted into numeric features before fitting a standard scikit-learn Naive Bayes classifier?

- A. The classifier expects a numerical feature representation
- B. The classifier can only predict numbers
- C. Text labels are forbidden
- D. All words must be removed

21. What is Laplace smoothing intended to address?

- A. Zero probabilities caused by unseen feature–class combinations
- B. Incorrect train–test splitting
- C. Missing target labels in all datasets
- D. Excessively deep decision trees

22. In MultinomialNB(alpha=1.0), what does alpha control?

- A. Additive smoothing
- B. Number of nearest neighbors
- C. Maximum tree depth
- D. Regularization strength C

23. What may happen without smoothing when a word has never appeared in training examples of a class?

- A. The estimated likelihood for that word under the class may be zero
- B. The word is automatically assigned to every class
- C. All training labels are deleted
- D. The model becomes a regression model

24. Which statement about the independence assumption is most accurate?

- A. It is an approximation that may still yield useful predictions
- B. It must be exactly true for the model to make any prediction
- C. It means features are independent of the target
- D. It means training samples cannot share features

25. Consider the words “high” and “fever” in medical notes. What might violate the naive assumption?

- A. Their occurrences may be related even after accounting for the class
- B. Both words contain letters
- C. One word is shorter than the other
- D. Both words can be converted into numbers

26. Which dataset is most naturally suited to GaussianNB among these choices?

- A. Continuous measurements such as temperature and blood pressure
- B. Raw text strings without numerical conversion
- C. Document word counts with thousands of tokens
- D. Binary word-presence features only

27. Does “Gaussian” in GaussianNB describe the assumed distribution of each continuous feature within each class?
- A. Yes
 - B. No, it specifies the number of classes
 - C. No, it specifies the train–test split
 - D. No, it specifies the evaluation metric
28. A model predicts “positive” with probability 0.8 and “negative” with probability 0.2. What is its usual predicted class?
- A. Positive
 - B. Negative
 - C. Both classes
 - D. Neither class
29. What does `model.classes_` show after fitting a scikit-learn classifier?
- A. The class labels and their probability-column order
 - B. The number of input features only
 - C. The feature importance scores
 - D. The original training samples
30. Why should a medical screening classifier be evaluated with more than accuracy?
- A. Missing positive cases and issuing false alarms can have different consequences
 - B. Accuracy cannot be computed for Naive Bayes
 - C. Every dataset must have exactly three classes
 - D. Naive Bayes always predicts one class
31. Which metric measures the fraction of actual positive cases identified?
- A. Recall
 - B. Specificity
 - C. Mean squared error
 - D. Tree depth
32. Which metric measures the fraction of predicted positive cases that are truly positive?
- A. Precision
 - B. Recall
 - C. Number of features
 - D. Training time
33. What can a confusion matrix show for a binary Naive Bayes classifier?
- A. Counts of correct and incorrect predictions by class
 - B. The distribution assumed for each feature
 - C. The exact words in every document
 - D. The source code of the model
34. Why should CountVectorizer be fitted on training documents rather than all documents before splitting?
- A. To avoid learning vocabulary information from the test set
 - B. To make the test set larger
 - C. To increase the number of classes
 - D. To guarantee 100% accuracy
35. What is a practical benefit of using a scikit-learn Pipeline for text classification?
- A. It keeps text transformation and classification together during training and evaluation
 - B. It removes the need for a target variable

- C. It ensures all predictions are correct
 - D. It converts classification into clustering
36. If a word occurs frequently in both spam and legitimate emails, what is likely true?
- A. That word alone may provide limited discrimination between the classes
 - B. The model must always classify the email as spam
 - C. The word will be deleted automatically
 - D. The training process must fail
37. If a word is common in spam but rare in legitimate emails, how might it affect prediction?
- A. Its presence can support the spam class
 - B. Its presence always makes both classes equally probable
 - C. It prevents any class prediction
 - D. It reduces the number of input documents
38. What is the distinction between MultinomialNB and BernoulliNB in a typical text application?
- A. The former can use token counts; the latter commonly uses binary presence indicators
 - B. The former is for regression; the latter is for clustering
 - C. The former needs images; the latter needs audio
 - D. They are both decision tree algorithms
39. Which target is suitable for Naive Bayes classification?
- A. "Spam" versus "not spam"
 - B. Exact house price in rupees
 - C. Exact patient height in centimeters
 - D. Continuous rainfall amount
40. What should you do if the training and deployment data have substantially different class proportions?
- A. Examine whether changed priors affect performance and probability estimates
 - B. Assume performance must be identical
 - C. Remove all class labels
 - D. Increase tree depth
41. What does an estimated probability of 0.9 from predict_proba require before being trusted as a real-world risk estimate?
- A. Calibration and evaluation on representative data
 - B. Only a larger Python font
 - C. Changing every feature to a string
 - D. Setting the test size to zero
42. Can Naive Bayes handle a classification problem with more than two classes?
- A. Yes
 - B. No, it is always binary
 - C. Only if K is odd
 - D. Only with a random forest
43. Which statement best describes the relationship between features and class in Naive Bayes?
- A. The model uses feature evidence to compare candidate classes
 - B. The model ignores all feature values
 - C. The model predicts the class from row order
 - D. The model requires one feature for each class

44. What does GaussianNB estimate from the training data for each feature and class?
- A. Parameters such as mean and variance
 - B. Only tree depth
 - C. Nearest-neighbor indices
 - D. A sequence of gradient updates
45. What is a limitation when strongly related features are included in Naive Bayes?
- A. Their evidence can effectively be counted more than intended
 - B. They are always merged automatically
 - C. The target labels are erased
 - D. The classifier cannot output probabilities
46. Why is a held-out test set useful?
- A. It estimates how the trained model performs on unseen examples
 - B. It is used to calculate training priors
 - C. It replaces feature extraction
 - D. It guarantees deployment performance
47. What does `classification_report(y_test, y_pred)` commonly display?
- A. Precision, recall, F1-score, and support
 - B. Only a histogram of feature values
 - C. The source code of the model
 - D. The optimal learning rate
48. A model marks every email as “not spam” in a dataset with very little spam. What should be checked carefully?
- A. Performance on the spam class
 - B. Only overall accuracy
 - C. Maximum tree depth
 - D. Euclidean distance
49. Which feature representation would be appropriate for using MultinomialNB on documents?
- A. Nonnegative token-count features
 - B. Raw Python strings passed directly as numeric arrays
 - C. Arbitrary negative word counts
 - D. Image file paths only
50. When would Naive Bayes be a reasonable first model?
- A. When building a fast, interpretable baseline for labeled text classification
 - B. When no input features exist
 - C. When the output must be an image
 - D. When every feature relationship must be modeled exactly

3. Decision Trees

1. How does a decision tree classifier make a prediction?
 - A. It follows feature-based splits from the root to a leaf
 - B. It averages distances to all samples
 - C. It multiplies word likelihoods only
 - D. It always predicts the most recent label
2. What does an internal node of a decision tree usually represent?
 - A. A test or split on a feature
 - B. The final prediction only
 - C. A missing dataset
 - D. A test accuracy value
3. What does a leaf node usually contain?
 - A. A prediction or class distribution
 - B. Another training dataset
 - C. A Python import statement
 - D. A distance metric
4. Which scikit-learn class builds a decision tree for classification?
 - A. DecisionTreeClassifier
 - B. DecisionTreeRegressor
 - C. KNeighborsClassifier
 - D. LogisticRegression
5. Which scikit-learn class builds a decision tree for continuous-value prediction?
 - A. DecisionTreeRegressor
 - B. DecisionTreeClassifier
 - C. GaussianNB
 - D. RandomForestClassifier
6. Which import statement is appropriate for tree classification?
 - A. `from sklearn.tree import DecisionTreeClassifier`
 - B. `from sklearn.ensemble import DecisionTreeClassifier`
 - C. `from sklearn.neighbors import DecisionTreeClassifier`
 - D. `from sklearn.metrics import DecisionTreeClassifier`
7. What does `model.fit(X_train, y_train)` do for a decision tree?
 - A. Learns splits from labeled training data
 - B. Draws the tree without learning
 - C. Predicts labels for test data only
 - D. Calculates the confusion matrix
8. What does `model.predict(X_test)` return for a decision tree classifier?
 - A. Predicted class labels
 - B. Only tree depth
 - C. Training samples
 - D. Feature names only
9. What is the root node?
 - A. The first split at the top of the tree
 - B. The final leaf on the left

- C. The test dataset
 - D. The most frequent target value
10. What is the purpose of selecting a split during tree training?
- A. To separate examples into groups that are more useful for prediction
 - B. To randomly rename features
 - C. To standardize every feature
 - D. To remove the target labels
11. In classification trees, what does Gini impurity measure?
- A. How mixed the class labels are within a node
 - B. The physical distance between records
 - C. The number of missing features
 - D. The speed of Python execution
12. A node contains examples from only one class. How would you describe it?
- A. Pure
 - B. Untrained
 - C. Standardized
 - D. Randomized
13. What does `criterion="entropy"` change in `DecisionTreeClassifier`?
- A. The measure used to evaluate candidate classification splits
 - B. The number of training rows
 - C. The target variable type
 - D. The Python package being imported
14. What does `max_depth=3` limit?
- A. How deep the tree can grow
 - B. The number of input features to exactly three
 - C. The number of training samples to three
 - D. The number of classes to three
15. Why might a very deep decision tree overfit?
- A. It can learn narrow patterns specific to the training samples
 - B. It cannot split the data
 - C. It is forced to predict one class
 - D. It always uses too few features
16. What is a likely symptom of overfitting?
- A. Very high training performance but much lower test performance
 - B. Identical performance on all datasets by definition
 - C. No training predictions
 - D. No leaf nodes
17. What does `min_samples_leaf` control?
- A. The minimum number of training samples required in a leaf
 - B. The maximum number of trees
 - C. The minimum number of features in the dataset
 - D. The test-set fraction
18. How can increasing `min_samples_leaf` affect a tree?
- A. It can prevent very small, specialized leaves
 - B. It always makes the tree deeper

- C. It turns the tree into KNN
- D. It changes every continuous feature into a category

19. What does `min_samples_split` control?

- A. The minimum number of samples required for a node to be considered for splitting
- B. The number of test sets
- C. The number of class labels
- D. The number of trees in an ensemble

20. Why is feature scaling usually less important for a standard decision tree than for KNN?

- A. Tree splits depend on feature thresholds rather than distances between whole samples
- B. Trees accept only standardized inputs
- C. Trees ignore numerical features
- D. Scaling always changes every tree prediction

21. A tree uses the rule `age <= 45`. What happens to a sample with age 40?

- A. It follows the branch for values satisfying the condition
- B. It is removed from the dataset
- C. It follows both branches equally
- D. It cannot be classified

22. What is a practical strength of a small decision tree?

- A. Its decision rules can often be inspected
- B. It always has the highest possible accuracy
- C. It needs no training data
- D. It automatically explains causation

23. Which function can visualize a fitted scikit-learn decision tree?

- A. `plot_tree`
- B. `train_test_split`
- C. `accuracy_score`
- D. `StandardScaler`

24. What does `feature_importances_` provide for a fitted tree?

- A. Impurity-based importance scores for input features
- B. Proof that a feature causes the outcome
- C. The original feature values
- D. Class probabilities for each test sample

25. Why should feature importance be interpreted cautiously?

- A. Importance in the fitted model does not establish a causal relationship
- B. It always sums to a negative number
- C. It is available only before fitting
- D. It reports the test accuracy of each class

26. Which is an appropriate decision tree classification task?

- A. Predicting whether a loan application belongs to an approval category
- B. Predicting the exact loan amount as a class label without defining categories
- C. Counting the number of files in a folder
- D. Displaying a spreadsheet

27. Which is an appropriate decision tree regression task?

- A. Predicting a house's sale price
- B. Classifying an email as spam

- C. Identifying a disease category
 - D. Predicting whether a transaction is fraudulent
28. What does a classification tree commonly predict at a leaf?
- A. A class based on training examples reaching that leaf
 - B. The Euclidean distance to the root
 - C. The number of Python packages installed
 - D. The maximum input value in the entire dataset
29. How are predicted class probabilities commonly derived at a tree leaf?
- A. From the class proportions of training samples in that leaf
 - B. From the number of features only
 - C. From the tree's plotting colors
 - D. From the order of test samples
30. What is pruning in the context of decision trees?
- A. Reducing tree complexity by removing or limiting branches
 - B. Adding a new feature to every node
 - C. Scaling every feature to zero
 - D. Converting the tree into a neural network
31. What does `ccp_alpha` relate to in scikit-learn decision trees?
- A. Cost-complexity pruning
 - B. Number of nearest neighbors
 - C. Naive Bayes smoothing
 - D. Logistic regression regularization C
32. Why might cross-validation be used when choosing `max_depth`?
- A. To compare generalization performance across candidate depths
 - B. To determine the number of class names
 - C. To avoid fitting a model
 - D. To turn a test set into training data
33. What is a drawback of a single decision tree?
- A. Small changes in training data can sometimes produce a different tree
 - B. It cannot predict categories
 - C. It requires all features to be images
 - D. It always takes hours to make one prediction
34. What does `random_state=42` help with when a tree algorithm uses randomness?
- A. Reproducibility of results across runs
 - B. Guaranteeing perfect performance
 - C. Increasing the number of labels
 - D. Replacing cross-validation
35. What should you examine if a tree predicts almost every sample as the majority class?
- A. Class imbalance and per-class evaluation metrics
 - B. Only the Python variable name
 - C. The screen resolution
 - D. Only the number of plotted nodes
36. What does `class_weight="balanced"` aim to adjust?
- A. The influence of classes based on their frequencies
 - B. The physical weight of the computer

- C. The number of input columns
 - D. The depth of every branch to the same value
37. Can a decision tree represent nonlinear relationships between inputs and output?
- A. Yes, through combinations of feature splits
 - B. No, every tree is a straight line
 - C. Only when K equals 1
 - D. Only when no labels are provided
38. Why may a tree with depth 1 be too simple for a complex task?
- A. It makes only one split before predicting
 - B. It has too many leaves
 - C. It memorizes every training example
 - D. It contains multiple forests
39. What is a decision stump?
- A. A decision tree with a single split
 - B. A tree with no target variable
 - C. A random forest with one hundred trees
 - D. A tree used only for regression
40. What does `model.get_depth()` report?
- A. The depth of the fitted tree
 - B. The number of training features
 - C. Test accuracy
 - D. The number of classes
41. What does `model.get_n_leaves()` report?
- A. The number of leaf nodes
 - B. The number of training samples
 - C. The number of feature columns
 - D. The number of incorrect predictions
42. Why should you keep a separate test set when tuning a decision tree?
- A. To obtain a final evaluation on data not used for tuning
 - B. To increase training accuracy automatically
 - C. To ensure the tree has only two leaves
 - D. To avoid calculating predictions
43. Which evaluation metric is appropriate for a decision tree regressor?
- A. Mean absolute error
 - B. Classification accuracy only
 - C. Precision for a positive class
 - D. Confusion matrix only
44. Which evaluation metric is appropriate for a decision tree classifier?
- A. F1-score
 - B. Mean absolute error only
 - C. Mean squared error only
 - D. R-squared only
45. What is data leakage when preparing a tree model?
- A. Information unavailable at prediction time enters training or evaluation
 - B. Data is stored in two CSV files

- C. The tree has many leaf nodes
- D. The model uses an integer feature

46. A hospital model uses “discharge diagnosis” to predict disease at admission. What is the concern?

- A. The feature contains information that would not yet be available at admission
- B. The feature is categorical
- C. The tree must have three levels
- D. The target must be continuous

47. Which parameter directly restricts the maximum number of leaf nodes?

- A. `max_leaf_nodes`
- B. `n_neighbors`
- C. `alpha` in MultinomialNB
- D. `n_estimators`

48. What does an axis-aligned split test in a standard scikit-learn decision tree?

- A. One feature against a threshold at a time
- B. All features against a single combined coefficient
- C. The target against the test accuracy
- D. Only the first training sample

49. Why might one-hot encoding be used before fitting a tree with categorical input columns in scikit-learn?

- A. Standard scikit-learn tree estimators require numeric feature arrays
- B. Trees require every feature to have mean zero
- C. One-hot encoding creates target labels automatically
- D. It guarantees the smallest possible tree

50. When is a small decision tree a useful choice?

- A. When clear, inspectable rules are valuable
- B. When no labeled examples are available
- C. When the model must ignore all features
- D. When the only goal is to store raw files

4. Random Forest

1. What is a random forest classifier?
 - A. An ensemble of decision trees used together for classification
 - B. A single logistic equation
 - C. A nearest-neighbor search method
 - D. A text tokenizer
2. Why are multiple trees combined in a random forest?
 - A. To make predictions more stable than a single tree in many settings
 - B. To eliminate the need for data
 - C. To ensure every tree is identical
 - D. To convert classification into clustering
3. Which scikit-learn class is used for random forest classification?
 - A. RandomForestClassifier
 - B. DecisionTreeRegressor
 - C. KNeighborsClassifier
 - D. GaussianNB
4. Which class is used to predict continuous values with a random forest?
 - A. RandomForestRegressor
 - B. RandomForestClassifier
 - C. LogisticRegression
 - D. BernoulliNB
5. Which import statement is appropriate for random forest classification?
 - A. `from sklearn.ensemble import RandomForestClassifier`
 - B. `from sklearn.tree import RandomForestClassifier`
 - C. `from sklearn.naive_bayes import RandomForestClassifier`
 - D. `from sklearn.neighbors import RandomForestClassifier`
6. What does `n_estimators=100` specify?
 - A. Build 100 trees
 - B. Use 100 input features
 - C. Create 100 target classes
 - D. Train for 100 epochs
7. How does a random forest commonly produce a classification prediction?
 - A. By combining the predictions of its trees
 - B. By using only the deepest tree
 - C. By finding the nearest training sample
 - D. By averaging the input feature values
8. How does a random forest regressor commonly combine tree outputs?
 - A. It averages their numerical predictions
 - B. It chooses the alphabetically first prediction
 - C. It multiplies all predicted values
 - D. It reports only the highest tree value
9. What is bootstrap sampling in a random forest?
 - A. Sampling training rows with replacement for individual trees
 - B. Deleting all repeated feature values

- C. Scaling features to a common range
 - D. Sampling only from the test set
10. What does “with replacement” mean?
- A. The same training row can be selected more than once in a sample
 - B. Every selected row is permanently removed
 - C. Test rows replace training rows
 - D. Missing values are replaced by zero
11. Why does a random forest consider random subsets of features at splits?
- A. To introduce diversity among trees
 - B. To make all trees identical
 - C. To avoid using target labels
 - D. To force one feature to determine every split
12. What is a key distinction between a single decision tree and a random forest?
- A. A forest combines multiple trees
 - B. A forest never uses feature splits
 - C. A single tree requires no labeled data
 - D. A forest can only perform regression
13. What does `model.fit(X_train, y_train)` do?
- A. Fits the forest using labeled training examples
 - B. Predicts test labels without training
 - C. Generates a confusion matrix only
 - D. Removes the target column automatically
14. What does `model.predict(X_test)` return from a forest classifier?
- A. Predicted class labels
 - B. Only the number of trees
 - C. Training indices
 - D. The names of all leaf nodes
15. What does `model.predict_proba(X_test)` provide?
- A. Estimated probabilities for each class
 - B. Only a list of feature names
 - C. Mean squared error
 - D. The bootstrap samples themselves
16. What does `random_state=42` help make reproducible?
- A. Random choices in fitting and their resulting model
 - B. The real-world data-generating process
 - C. Perfect test accuracy
 - D. The number of input features
17. Why is feature scaling usually not essential for a standard random forest?
- A. Tree splits rely on thresholds rather than distances between full samples
 - B. The forest scales all features automatically to mean zero
 - C. Forests use only categorical features
 - D. Forests cannot accept scaled data
18. What is an out-of-bag (OOB) sample for a particular tree?
- A. A training sample not selected for that tree’s bootstrap sample
 - B. A sample from the final test set

- C. A sample with a missing label
- D. A sample with only one feature

19. What does `oob_score=True` enable when bootstrap sampling is used?

- A. An internal performance estimate using out-of-bag training samples
- B. Automatic removal of all outliers
- C. A guarantee of unbiased real-world performance
- D. A switch to logistic regression

20. Should OOB performance automatically replace a truly independent final test evaluation?

- A. No
- B. Yes, in all cases
- C. Yes, but only for regression
- D. Yes, if there are more than ten trees

21. What does `max_depth` control in a random forest?

- A. The maximum depth of its individual trees
- B. The number of forests
- C. The number of classes
- D. The size of the test set

22. What can increasing `min_samples_leaf` do?

- A. Make individual trees less focused on tiny groups of training samples
- B. Force every tree to have more depth
- C. Set the number of trees to one
- D. Remove all continuous features

23. What does `max_features` influence?

- A. How many features are considered when looking for a split
- B. How many target labels can be predicted
- C. How many predictions can be made
- D. How many CSV files can be loaded

24. What is a likely result if all trees are nearly identical?

- A. Combining them offers less benefit from diversity
- B. Their votes become invalid Python syntax
- C. The number of classes doubles
- D. The forest becomes an unsupervised model

25. Which task is suitable for `RandomForestClassifier`?

- A. Predicting whether a transaction is fraudulent
- B. Predicting an exact continuous sales amount
- C. Compressing an image file
- D. Sorting patient IDs

26. Which task is suitable for `RandomForestRegressor`?

- A. Predicting a patient's length of stay in days as a numerical value
- B. Assigning a document to a topic category
- C. Classifying a transaction as fraudulent
- D. Identifying an email as spam

27. What is one reason a random forest may outperform a single deep tree on unseen data?

- A. Combining varied trees can reduce sensitivity to individual training examples
- B. Every tree is guaranteed to be perfectly accurate

- C. It removes the need for a training set
 - D. It uses the test labels during fitting
28. What is a disadvantage of a forest compared with a small single tree?
- A. Its overall prediction process is harder to explain as a short set of rules
 - B. It cannot use numeric features
 - C. It cannot make classification predictions
 - D. It has no adjustable parameters
29. What does `feature_importances_` report in a standard fitted scikit-learn forest?
- A. Impurity-based feature importance scores
 - B. Proof of cause and effect
 - C. Predictions for the test set
 - D. The number of missing values per feature
30. Why can impurity-based feature importance be misleading?
- A. It can favor certain feature types and does not establish causation
 - B. It is always identical for every feature
 - C. It is calculated from test labels only
 - D. It cannot be computed for forests
31. What is permutation importance based on?
- A. Measuring performance changes after shuffling a feature's values
 - B. Counting the letters in feature names
 - C. Deleting the target labels
 - D. Replacing all trees with one tree
32. Why should feature importance ideally be examined on suitable held-out data?
- A. To see whether the feature helps predictions beyond the training examples
 - B. To force all importance values to be equal
 - C. To avoid making any predictions
 - D. To increase the number of classes
33. What does `class_weight="balanced"` address in a forest classifier?
- A. Differences in class frequency
 - B. Different physical units of features
 - C. Different file sizes of datasets
 - D. Different depths of plotted trees only
34. A fraud dataset contains 99% legitimate transactions. Which evaluation is especially useful?
- A. Precision and recall for fraud
 - B. Training accuracy alone
 - C. Number of trees alone
 - D. Feature count alone
35. Why can overall accuracy be misleading in that fraud dataset?
- A. A model can predict "legitimate" almost always and still achieve high accuracy
 - B. Accuracy cannot be calculated for forests
 - C. Fraud must be a continuous target
 - D. Forests ignore class imbalance by definition
36. What may happen as `n_estimators` increases substantially?
- A. Fitting and prediction may require more computation
 - B. The target becomes continuous

- C. The dataset gains new observations automatically
 - D. Every tree becomes shallower by definition
37. Does adding more trees always guarantee better test performance?
- A. No
 - B. Yes
 - C. Yes, but only when there are two features
 - D. Yes, if the training accuracy is high
38. What does `n_jobs=-1` commonly request in a scikit-learn forest?
- A. Use all available CPU workers for supported parallel computation
 - B. Train one less tree than requested
 - C. Remove the last feature
 - D. Use GPU acceleration automatically
39. Why should you use cross-validation when selecting forest hyperparameters?
- A. To compare candidate settings on multiple training/validation splits
 - B. To obtain the true labels for future samples
 - C. To guarantee a specific test score
 - D. To eliminate the need for training
40. Which code correctly constructs and fits a forest classifier?
- A. `model = RandomForestClassifier(n_estimators=100); model.fit(X_train, y_train)`
 - B. `model = RandomForestClassifier.fit(X_test)`
 - C. `model = RandomForestClassifier(y_train); model.predict(X_train, y_train)`
 - D. `model = fit(RandomForestClassifier, X_test)`
41. What does `train_test_split(..., stratify=y)` help preserve in classification?
- A. Approximate class proportions across splits
 - B. Feature scaling parameters
 - C. Identical rows in both splits
 - D. The same prediction for every class
42. What is data leakage in a forest-based project?
- A. Training or model selection uses information that would not be available at prediction time
 - B. Using more than one tree
 - C. Using a categorical target
 - D. Setting `random_state`
43. A hospital dataset includes an outcome recorded after treatment, but the model is meant to predict at admission. What should you do?
- A. Exclude that future-information feature from admission-time prediction
 - B. Use it because it improves training accuracy
 - C. Duplicate it across several columns
 - D. Increase `n_estimators` until it disappears
44. Can a random forest model interactions between features?
- A. Yes, tree paths can combine splits on multiple features
 - B. No, every feature is considered completely independently
 - C. Only if all features are standardized
 - D. Only when there are exactly two classes
45. How does a forest typically behave when asked to predict a regression value far outside the range of observed target values?

- A. It generally has limited ability to extrapolate beyond observed target ranges
 - B. It always extrapolates perfectly
 - C. It changes into a linear model
 - D. It refuses to predict any value
46. Why might one-hot encoding be used for categorical input features with scikit-learn random forests?
- A. The estimator expects numeric feature input
 - B. It forces every tree to have exactly one leaf
 - C. It guarantees feature importance is causal
 - D. It creates the target variable
47. What is a reasonable way to compare a decision tree and a random forest?
- A. Evaluate both on the same appropriate validation or test data and metrics
 - B. Compare only the length of their Python class names
 - C. Compare training accuracy only
 - D. Assume the forest always wins
48. What does a random forest's predicted probability of 0.8 require before use as a reliable risk estimate?
- A. Evaluation and possibly calibration on representative data
 - B. More trees alone, with no evaluation
 - C. Converting every feature to a string
 - D. A deeper single tree
49. What is the forest's use of randomness intended to help create?
- A. Diversity among its trees
 - B. Random target labels for test samples
 - C. Missing values in every feature
 - D. An identical tree in every position
50. When is a random forest a practical baseline?
- A. For labeled tabular data with potentially nonlinear relationships
 - B. When the task has no inputs and no target
 - C. When the only goal is to edit images
 - D. When a single fixed linear coefficient is required for every feature

5. Logistic Regression

1. Despite its name, what is logistic regression commonly used for?
 - A. Classification
 - B. Image compression
 - C. Clustering only
 - D. Sorting rows
2. In binary logistic regression, what is commonly predicted before applying a decision threshold?
 - A. A probability-like estimate for a class
 - B. The nearest training sample
 - C. The depth of a tree
 - D. The number of clusters
3. Which scikit-learn class implements logistic regression?
 - A. LogisticRegression
 - B. LinearRegression
 - C. LogisticClassifierTree
 - D. GaussianNB
4. Which import statement is appropriate?
 - A. `from sklearn.linear_model import LogisticRegression`
 - B. `from sklearn.tree import LogisticRegression`
 - C. `from sklearn.neighbors import LogisticRegression`
 - D. `from sklearn.metrics import LogisticRegression`
5. Which is a suitable binary logistic regression target?
 - A. Disease present or absent
 - B. Exact temperature in degrees Celsius
 - C. Exact product price
 - D. Exact patient weight
6. What does `model.fit(X_train, y_train)` do?
 - A. Estimates model parameters using labeled training data
 - B. Returns test accuracy without training
 - C. Converts the model into KNN
 - D. Creates a decision tree plot
7. What does `model.predict(X_test)` return?
 - A. Predicted class labels
 - B. Only coefficient values
 - C. The input feature names
 - D. The training data
8. What does `model.predict_proba(X_test)` return?
 - A. Estimated probabilities for each class
 - B. Only the confusion matrix
 - C. Euclidean distances
 - D. Training-set indices
9. What function maps a real-valued score into a value between 0 and 1 in binary logistic regression?
 - A. Sigmoid function
 - B. Gini function

- C. Euclidean function
- D. Bootstrap function

10. What is a decision threshold?

- A. A probability cutoff used to convert a score into a class prediction
- B. The number of training features
- C. The number of model coefficients
- D. The number of validation folds

11. With a 0.5 threshold, what class is usually assigned when the estimated positive-class probability is 0.8?

- A. Positive
- B. Negative
- C. Both classes
- D. No class

12. Why might a medical screening system use a threshold below 0.5?

- A. To identify more potential positive cases, while potentially increasing false alarms
- B. To remove all false positives and false negatives
- C. To make the model a random forest
- D. To eliminate the need for validation

13. What generally happens to recall for the positive class when the positive decision threshold is lowered?

- A. It may increase
- B. It must become zero
- C. It always remains identical
- D. It becomes undefined for every dataset

14. What generally happens to the number of false positives when the positive threshold is lowered?

- A. It may increase
- B. It must become zero
- C. It always remains identical
- D. It becomes equal to the number of features

15. What does a positive coefficient suggest, with other features held fixed?

- A. Larger values of that feature increase the model's log-odds of the positive class
- B. Larger values always decrease predicted probability
- C. The feature is necessarily causal
- D. The feature is ignored

16. Does a larger positive coefficient always mean a feature is more important than another feature measured on a different scale?

- A. No, feature units and scaling affect coefficient magnitudes
- B. Yes, regardless of units
- C. Yes, but only if there are three classes
- D. No, because coefficients cannot be positive

17. Why is feature scaling often helpful for logistic regression?

- A. It can improve optimization and make regularization act more comparably across features
- B. It converts classification into regression
- C. It guarantees perfect calibration
- D. It changes the number of labels

18. Which preprocessing tool is commonly used to standardize numeric features?
- A. StandardScaler
 - B. CountVectorizer for all numeric data
 - C. plot_tree
 - D. confusion_matrix
19. Which code structure combines scaling and logistic regression appropriately?
- A. Pipeline([("scaler", StandardScaler()), ("model", LogisticRegression())])
 - B. Pipeline([("model", LogisticRegression()), ("scaler", StandardScaler())])
 - C. LogisticRegression(StandardScaler())
 - D. StandardScaler.fit(LogisticRegression())
20. Why should preprocessing be fitted within training folds during cross-validation?
- A. To prevent validation information leaking into training
 - B. To ensure the validation set has no rows
 - C. To increase the number of classes
 - D. To make all coefficients equal
21. What does regularization generally do?
- A. Discourages excessively large coefficients
 - B. Adds more target classes
 - C. Selects the nearest neighbors
 - D. Creates deeper tree branches
22. In scikit-learn logistic regression, what does a smaller C generally mean?
- A. Stronger regularization
 - B. Weaker regularization
 - C. More trees
 - D. More nearest neighbors
23. What does a larger C generally mean?
- A. Weaker regularization
 - B. Stronger regularization
 - C. Fewer training rows
 - D. More classes by definition
24. What can L1 regularization encourage?
- A. Some coefficients becoming exactly zero
 - B. More trees in an ensemble
 - C. Every feature becoming identical
 - D. All probabilities becoming 0.5
25. What does L2 regularization typically encourage?
- A. Smaller coefficients without generally forcing them all to zero
 - B. A tree with two leaves
 - C. Equal class frequencies
 - D. A fixed number of neighbors
26. What is the decision boundary of basic binary logistic regression in the original feature space?
- A. Linear in the supplied features
 - B. Always circular
 - C. A random collection of tree branches
 - D. Based only on the nearest sample

27. Can logistic regression represent a nonlinear pattern if suitable transformed features are supplied?
- A. Yes
 - B. No, under any circumstances
 - C. Only with bootstrap sampling
 - D. Only if labels are missing
28. Which transformation could help model a curved relationship?
- A. Adding polynomial or interaction features
 - B. Removing every input feature
 - C. Replacing the target with row numbers
 - D. Increasing the test set to 100%
29. What are log-odds in logistic regression?
- A. The logarithm of the odds of the positive class
 - B. The logarithm of the number of features
 - C. The number of false positives
 - D. The distance between training samples
30. If a coefficient is zero, what does that mean within the fitted linear score?
- A. That feature makes no direct contribution to the score through that coefficient
 - B. The feature must be missing from the original dataset
 - C. The target is always zero
 - D. The model cannot predict probabilities
31. Which metric measures how many predicted positives are actually positive?
- A. Precision
 - B. Recall
 - C. Mean squared error
 - D. R-squared
32. Which metric measures how many actual positives are identified?
- A. Recall
 - B. Precision
 - C. Number of coefficients
 - D. Training sample count
33. Why may accuracy be misleading for rare-event classification?
- A. Predicting the majority class can give high accuracy while missing rare events
 - B. Accuracy cannot be used for logistic regression
 - C. Accuracy always exceeds 1
 - D. It measures tree depth
34. What does a confusion matrix summarize?
- A. Counts of correct and incorrect predictions by actual and predicted class
 - B. The values of model coefficients only
 - C. The feature scaling means only
 - D. The number of optimization iterations only
35. What does ROC AUC evaluate?
- A. How well model scores rank positive examples above negative examples across thresholds
 - B. Only performance at exactly one fixed threshold
 - C. Mean absolute error
 - D. The number of input features

36. What can a precision–recall curve be especially informative for?
- A. Evaluating positive-class detection in an imbalanced dataset
 - B. Selecting the number of trees
 - C. Measuring tree depth
 - D. Computing Euclidean distance
37. Why should predicted probabilities be checked for calibration?
- A. A predicted risk of 0.8 should correspond reasonably to observed outcomes near that level
 - B. Calibration changes the target from categorical to continuous
 - C. Calibration guarantees zero classification errors
 - D. Calibration is the same as feature scaling
38. Which scikit-learn attribute contains fitted coefficients?
- A. `coef_`
 - B. `feature_importances_`
 - C. `n_neighbors_`
 - D. `tree_`
39. Which attribute contains the fitted intercept?
- A. `intercept_`
 - B. `max_depth_`
 - C. `bootstrap_`
 - D. `classes_count_`
40. What does `model.classes_` help you identify?
- A. Which class corresponds to each column of `predict_proba`
 - B. The order of training rows
 - C. The names of numerical features
 - D. The number of validation folds
41. Can scikit-learn logistic regression handle multiclass classification?
- A. Yes
 - B. No, only two classes are possible
 - C. Only when K is odd
 - D. Only when features are text strings
42. Which is an example of multiclass classification?
- A. Assigning a document to sports, politics, or science
 - B. Predicting exact monthly revenue
 - C. Predicting exact blood glucose level
 - D. Estimating the number of trees
43. What is the purpose of `class_weight="balanced"`?
- A. To weight classes inversely according to their frequency
 - B. To scale each input feature to the same range
 - C. To make every coefficient positive
 - D. To set the decision threshold to exactly 0.8
44. Why should you avoid fitting a logistic regression model using a feature recorded after the outcome?
- A. It may leak information unavailable at prediction time
 - B. Logistic regression accepts only earlier dates as numbers
 - C. The number of classes will change
 - D. It creates too many trees

45. What does convergence mean during model fitting?
- A. The optimization procedure has reached its stopping condition for parameter estimation
 - B. All predictions are correct
 - C. Every coefficient is zero
 - D. The dataset has become balanced
46. If scikit-learn warns that logistic regression failed to converge, what is a sensible first check?
- A. Feature scaling, solver settings, and the iteration limit
 - B. Whether the variable name contains “logistic”
 - C. Whether `n_estimators` is at least 100
 - D. Whether `K` is an odd number
47. What does `max_iter` control?
- A. The maximum number of optimization iterations
 - B. The number of nearest neighbors
 - C. The maximum tree depth
 - D. The number of training features
48. Why is a separate test set useful after tuning `C` and other choices?
- A. It provides a final assessment on examples not used to choose the model
 - B. It increases the number of model coefficients
 - C. It removes the need for training labels
 - D. It guarantees perfect deployment performance
49. When is logistic regression a sensible baseline?
- A. When you want a relatively simple classifier with inspectable coefficients and predicted probabilities
 - B. When no labeled data are available
 - C. When the output must be a decision tree diagram
 - D. When the task is to find nearest images without labels
50. A model predicts a patient’s positive-class probability as 0.72. What should determine whether this produces a positive classification?
- A. The chosen decision threshold
 - B. The patient’s row number
 - C. The number of trees in the dataset
 - D. The alphabetical order of features